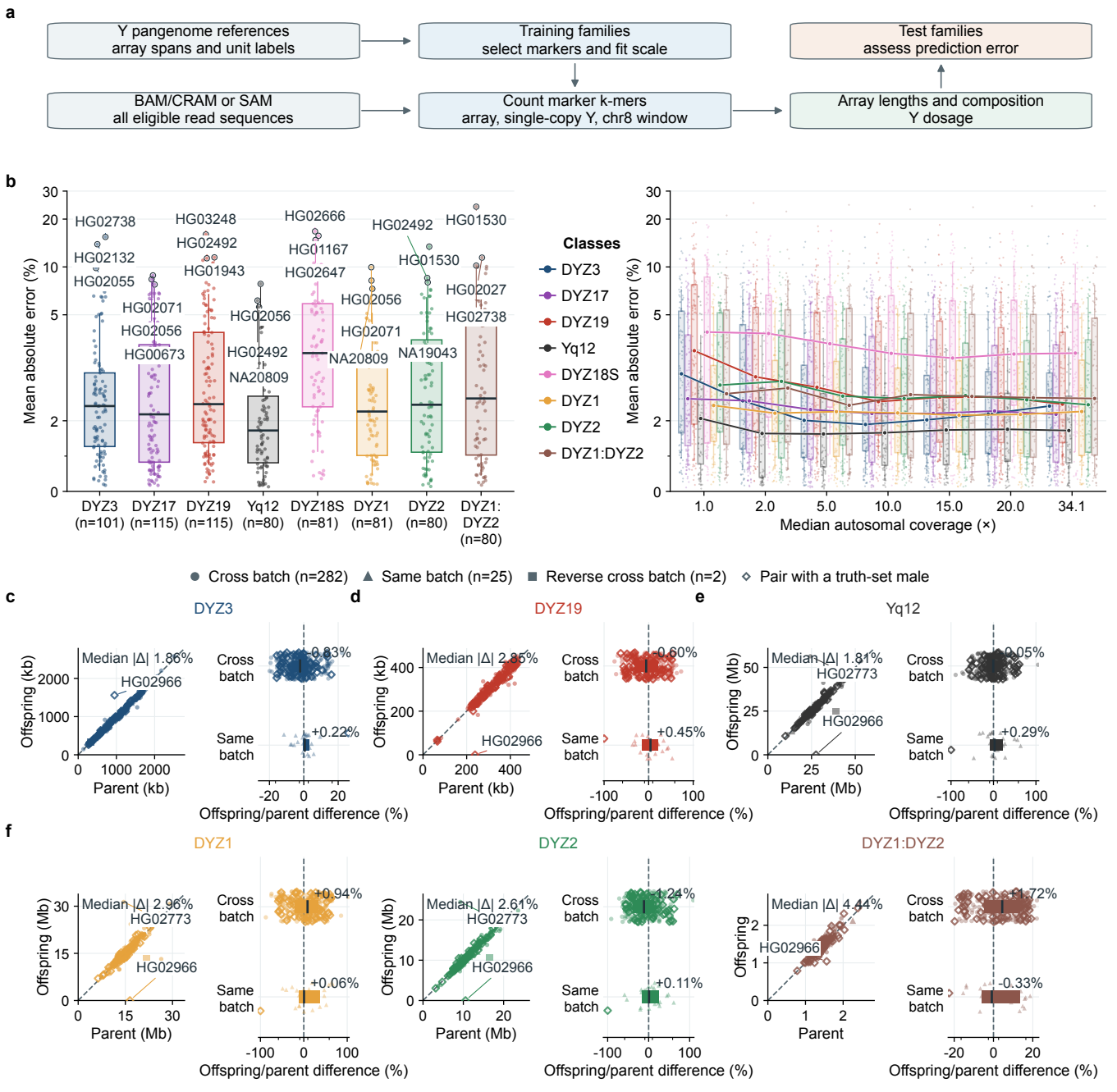

a
Y pangenome references
array spans and unit labels
Training families
select markers and fit scale
Test families
assess prediction error
BAM/CRAM or SAM
all eligible read sequences
Count marker k-mers
array, single-copy Y, chr8 window
Array lengths and composition
Y dosage
b
Mean absolute error (%)
DYZ3 (n=101)
DYZ17 (n=115)
DYZ19 (n=115)
Yq12 (n=80)
DYZ18S (n=81)
DYZ1 (n=81)
DYZ2 (n=80)
DYZ1: DYZ2 (n=80)
Classes
DYZ3
DYZ17
DYZ19
Yq12
DYZ18S
DYZ1
DYZ2
DYZ1:DYZ2
Median autosomal coverage (×)
Cross batch (n=282)
Same batch (n=25)
Reverse cross batch (n=2)
Pair with a truth-set male
c
DYZ3
Median |Δ| 1.86%
HG02966
Offspring (kb)
Parent (kb)
Cross batch
Same batch
0.83%
+0.22%
Offspring/parent difference (%)
d
DYZ19
Median |Δ| 2.85%
HG02966
Offspring (kb)
Parent (kb)
Cross batch
Same batch
0.60%
+0.45%
Offspring/parent difference (%)
e
Yq12
Median |Δ| 1.81%
HG02773
HG02966
Offspring (Mb)
Parent (Mb)
Cross batch
Same batch
0.05%
+0.29%
Offspring/parent difference (%)
f
DYZ1
Median |Δ| 2.96%
HG02773
HG02966
Offspring (Mb)
Parent (Mb)
Cross batch
Same batch
+0.94%
+0.06%
Offspring/parent difference (%)
DYZ2
Median |Δ| 2.61%
HG02773
HG02966
Offspring (Mb)
Parent (Mb)
Cross batch
Same batch
1.24%
+0.11%
Offspring/parent difference (%)
DYZ1:DYZ2
Median |Δ| 4.44%
HG02966
Offspring
Parent
Cross batch
Same batch
1.72%
-0.33%
Offspring/parent difference (%)